

CHAPTER 8

WORK, ENERGY & POWER

1.7 Energy, work and power

1.7.1 Energy

- 1 State that energy may be stored as kinetic, gravitational potential, chemical, elastic (strain), nuclear, electrostatic and internal (thermal)
- 2 Describe how energy is transferred between stores during events and processes, including examples of transfer by forces (mechanical work done), electrical currents (electrical work done), heating, and by electromagnetic, sound and other waves
- 3 Know the principle of the conservation of energy and apply this principle to the transfer of energy between stores during events and processes
- 4 Recall and use the equation for kinetic energy

$$E_k = \frac{1}{2}mv^2$$

- 5 Recall and use the equation for the change in gravitational potential energy

$$\Delta E_p = mg\Delta h$$

1.7.2 Work

- 1 Recall and use the equation
work done = force $\times$ distance moved in the direction of the force
 $W = Fd$

1.7.3 Energy resources

- 1 List renewable and non-renewable energy sources
- 2 Describe how useful energy may be obtained, or electrical power generated, from:
 - (a) chemical energy stored in fossil fuels
 - (b) chemical energy stored in biofuels
 - (c) hydroelectric resources
 - (d) solar radiation
 - (e) nuclear fuel
 - (f) geothermal resources
 - (g) wind
 - (h) tides
 - (i) waves in the seaincluding references to a boiler, turbine and generator where they are used
- 3 Describe advantages and disadvantages of each method limited to whether it is renewable, when and whether it is available, and its impact on the environment

1.7.4 Efficiency

- 1 Define efficiency as:
 - (a) $(\%) \text{ efficiency} = \frac{(\text{useful energy output})}{(\text{total energy input})} (\times 100\%)$
 - (b) $(\%) \text{ efficiency} = \frac{(\text{useful power output})}{(\text{total power input})} (\times 100\%)$and recall and use these equations

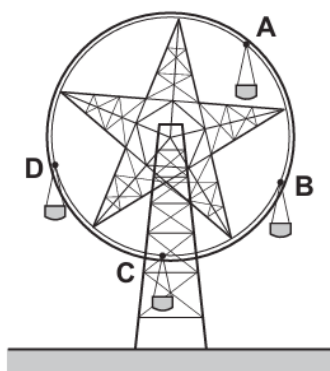
1.7.5 Power

- 1 Define power as work done per unit time and also as energy transferred per unit time; recall and use the equations
 - (a) $\text{power} = \frac{\text{work done}}{\text{time taken}}$
$$P = \frac{W}{t}$$
 - (b) $\text{power} = \frac{\text{energy transferred}}{\text{time taken}}$
$$P = \frac{\Delta E}{t}$$

1. O/N/23/11/Q8

The diagram shows a stationary fairground ride with four chairs of equal mass.

Which chair has the most gravitational potential energy?



2. O/N/23/11/Q9

Hydroelectric and tidal power stations generate electrical energy.

Do these use renewable sources of energy?

	hydroelectric	tidal
A	no	no
B	no	yes
C	yes	no
D	yes	yes

3. O/N/23/11/Q10

Which energy resource is correctly described?

	energy resource	renewable	availability
A	biofuels	yes	at all times
B	nuclear	yes	intermittent
C	solar radiation	no	at all times
D	wind	no	intermittent

4. O/N/23/11/Q11

A small motor has an input power rating of 10 W and is switched on for 5.0 minutes.

What is the electrical energy input to the motor in this time?

- A** 2.0 J **B** 50 J **C** 300 J **D** 3000 J

5. O/N/23/12/Q11

A box with mass 4.0 kg is pulled from rest along a horizontal floor by a constant force F .

A constant frictional force of 1.0 N acts on the box.

When the box has travelled 1.0 m its velocity is 1.5 m/s .

The kinetic energy of the box is equal to the work done on the box.

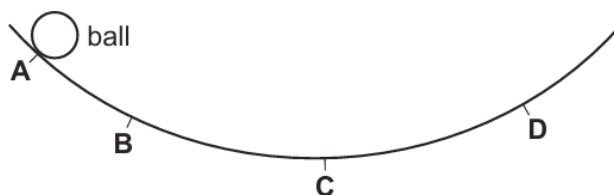
What is force F ?

- A** 2.0 N **B** 3.5 N **C** 4.0 N **D** 5.5 N

6. O/N/23/11/Q12

The diagram shows a curved track. A ball is released from the position shown.

At which point is the gravitational potential energy stored in the ball the maximum?



7. O/N/23/12/Q13

How does an oil-fired power station differ from a nuclear power station?

- A** Gases emitted by hot fuel are emitted into the atmosphere.
B Steam is produced in a boiler using hot fuel.
C The hot steam is used to turn a turbine.
D Turbines are used to drive an electric generator.

8. O/N/23/12/Q15

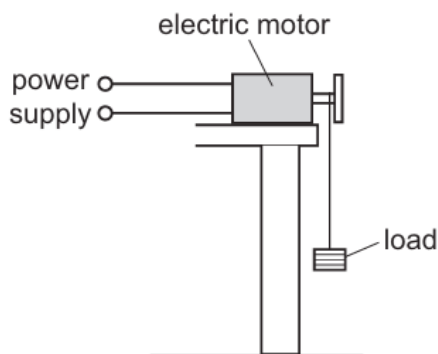
A battery delivers 3000 J of energy to a small 2.0 W electric motor before the battery stops working.

For how long is the motor switched on?

- A** 1500 minutes
B 100 minutes
C 50 minutes
D 25 minutes

9. O/N/23/12/Q14

An electric motor is used to lift a load at a constant speed.



The gravitational potential energy (GPE) stored in the load increases.

The moving load also has stored kinetic energy (KE).

Which expression is used to calculate the efficiency of the motor when the load is being lifted at a constant rate?

A $\text{efficiency} = \frac{\text{GPE gained per second}}{\text{KE gained per second}}$

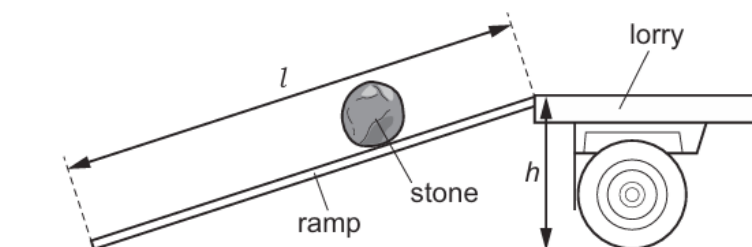
B $\text{efficiency} = \frac{\text{GPE gained per second}}{\text{power supplied}}$

C $\text{efficiency} = \frac{\text{power supplied}}{\text{GPE gained per second}}$

D $\text{efficiency} = \frac{\text{power supplied}}{\text{KE gained per second}}$

10. M/J/23/11/Q11

A large stone is rolled onto a lorry using a ramp.



The length of the ramp is l . The stone is lifted a height of h .

The mass of the stone is m . The weight of the stone is W .

Which expression is equal to the useful work done on the stone?

A $m \times h$

B $W \times h$

C $m \times l$

D $W \times l$

11. M/J/23/11/Q12

The diagram shows a burning candle.



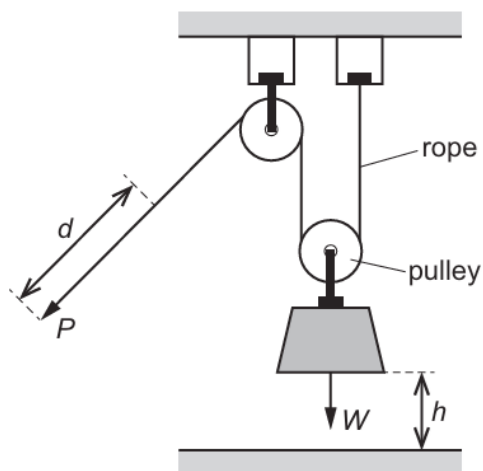
Which energy store is associated with the burning candle?

- A** chemical
- B** elastic
- C** electrostatic
- D** nuclear

12. M/J/23/12/Q12

A worker pulls a rope with a force P through a distance d in the direction of the force as shown.

This causes a block of weight W to move vertically upwards through a height h .



How much work is done by the worker?

- A** Ph
- B** Pd
- C** $(P + W)h$
- D** $(P + W)d$

13. M/J/23/12/Q13

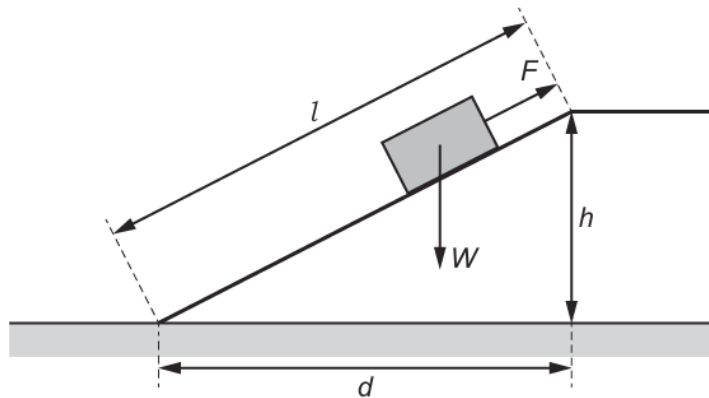
A battery is connected in a circuit to light a lamp.

What is the energy store in the battery?

- A** chemical
- B** electrostatic
- C** gravitational
- D** kinetic

14. O/N/22/11/Q15

A constant force F pulls a block of weight W up the slope shown.



How much work is done by F in pulling the block up the slope?

- A** $F \times h$ **B** $F \times l$ **C** $W \times d$ **D** $W \times l$

15. O/N/22/11/Q16

X and Y are both 40 W, 240 V lamps. Lamp X is more efficient than lamp Y.

Which statement correctly compares X with Y?

- A** X is more expensive to run.
B X has a lower electrical resistance.
C X produces less light energy.
D X produces less thermal energy.

16. O/N/22/12/Q17

The input power to a motor is 12 W.

The motor wastes 590 J of energy in 1.0 min.

What is the efficiency of the motor?

- A** 18% **B** 22% **C** 55% **D** 82%

17. O/N/22/12/Q18

Electric motors have an efficiency of about 90% when used in an electric train.

Which forms of wasted energy are produced?

- A** sound only
B thermal only
C thermal and chemical
D thermal and sound

18. O/N/22/12/Q19

What is a unit of power?

- A** J s **B** J/s **C** N s **D** N/s

19. M/J/22/11/Q8

A stone falls at terminal velocity in air.

What is equal to the change in gravitational potential energy of the stone?

- A** the change in kinetic energy
B the change in temperature of the stone
C the friction between the stone and the air
D the thermal energy produced

20. M/J/22/11/Q9

Water in a river is used to turn a turbine as the water flows downstream.

How is this type of power source described?

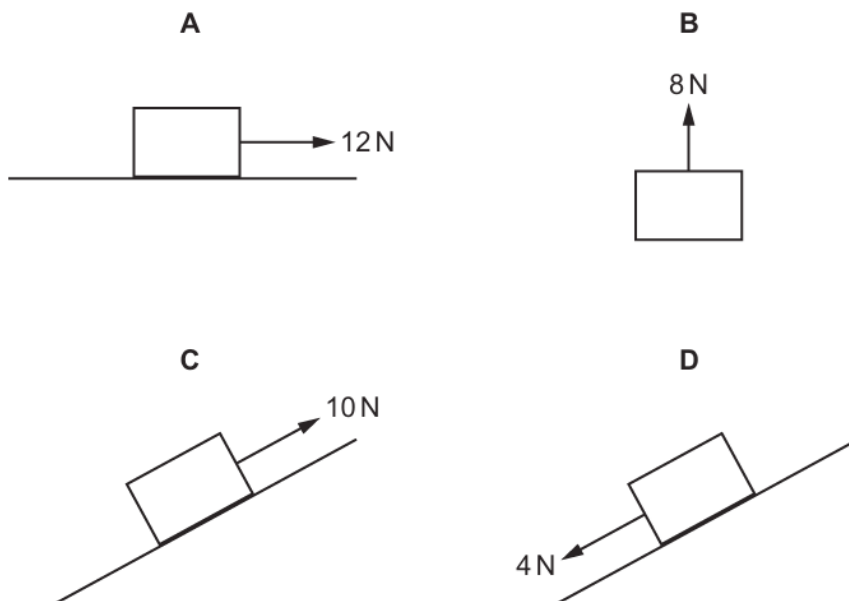
- A** hydroelectric
B tidal
C geothermal
D nuclear

21. M/J/22/11/Q10

A load is pulled by a rope attached to a motor. The resultant force exerted by the rope on the load is shown in the diagrams.

In each diagram, the load moves in the direction of the force shown and takes 10 s to travel 1.0 m.

In which diagram does the motor work with the greatest power?



22. M/J/22/11/Q11

The total energy incident in 1.0 s on a group of solar panels is 120 J/m^2 .

The group of solar panels converts 720 J of light energy to 120 J of useful electrical energy in 1.0 s.

What is the total surface area of the panels and what is the efficiency of the system?

	total surface area of solar panels / m^2	efficiency / %
A	6.0	17
B	0.17	6.0
C	6.0	6.0
D	0.17	17

23. M/J/22/11/Q12

A 1000 W, 240 V electrical device is switched on.

How much energy does it use in 5.0 minutes?

- A** 5000 J **B** 12000 J **C** 72000 J **D** 300000 J

24. M/J/22/12/Q13

Which device provides a continuous, steady energy output for the longest time?

- A** a nuclear reactor
B a solar panel
C a 1.5 V cell
D a wind generator

25. M/J/22/12/Q15

The electrical energy supplied to a kettle is 600 kJ.

Of this energy, 45 kJ is transferred to the surroundings and 15 kJ is used to heat the casing of the kettle.

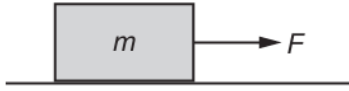
The remaining energy is used to heat the water.

What is the efficiency of the kettle?

- A** 0.10 **B** 0.90 **C** 0.925 **D** 0.975

26. O/N/21/11/Q11

The diagram shows a block of mass m pulled in a straight line along a horizontal surface by a force F .



The block moves a distance d in time t . The average speed at which the block moves is v .

Which two quantities must be known to calculate the work done?

- A** F and d **B** F and m **C** F and t **D** F and v

27. O/N/21/12/Q12

A 60 kg ice-skater, travelling initially at 3.0 m/s, glides across a horizontal icy surface. The skater brakes and comes to rest after 5.0 m.

The work done against the braking force is the change in the kinetic energy of the skater.

What is the braking force?

- A** 18 N **B** 27 N **C** 54 N **D** 108 N

28. O/N/21/12/Q13

Which equation is used to find the efficiency of an energy transfer?

A $\text{efficiency} = \frac{\text{useful energy output}}{\text{total energy input}}$

B $\text{efficiency} = \frac{\text{total energy input}}{\text{useful energy output}}$

C $\text{efficiency} = \frac{\text{energy converted to the required form}}{\text{total power input}}$

D $\text{efficiency} = (\text{energy converted to the required form}) \times (\text{total energy input})$

29. M/J/21/11/Q13

The work done by a force F on a body is calculated by multiplying F by a quantity q .

What is q ?

- A** the distance travelled in the direction of the force
B the distance travelled perpendicular to the direction of the force
C the velocity in the direction of the force
D the velocity in the direction perpendicular to the force

30. M/J/21/11/Q14

Some solar panels have a total area of 12 m^2 .

Each 1.0 m^2 of the panels receives 0.85 kJ of energy from the Sun in 1.0 s .

The efficiency of the panels is 16% .

How much power do they produce?

- A** 1.6 kW **B** 2.2 kW **C** 64 kW **D** 160 kW

31. M/J/21/12/Q16

A girl lifts an object of mass 1.0 kg from the floor and puts it on a ledge 2.0 m above the floor.

How much potential energy is gained by the object?

[gravitational field strength $g = 10 \text{ N/kg}$]

- A** 2.0 J **B** 5.0 J **C** 10 J **D** 20 J

32. O/N/20/11/O14

There is a current in a resistor.

Which energy transfer takes place?

- A** electrical $\rightarrow$ chemical
B electrical $\rightarrow$ internal
C kinetic $\rightarrow$ electrical
D potential $\rightarrow$ electrical

33. O/N/20/11/Q16

Which source of electrical energy is **not** renewable?

- A** solar cells
B hydroelectric generators
C wind turbines
D nuclear reactors

34. O/N/20/11/Q17

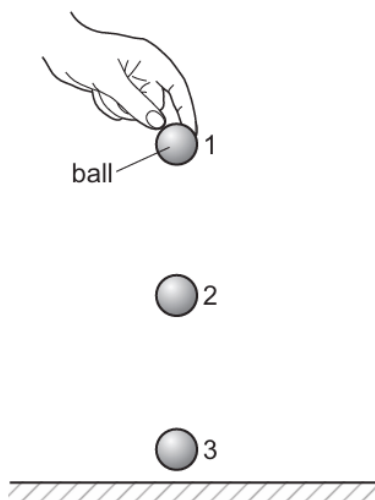
The table shows some data from a high-jump competition.

Which athlete jumps the highest?

	weight / N	increase in gravitational potential energy / J
A	600	1320
B	700	1610
C	800	1760
D	900	1800

35. O/N/20/11/Q15

A ball is dropped from a height as shown.

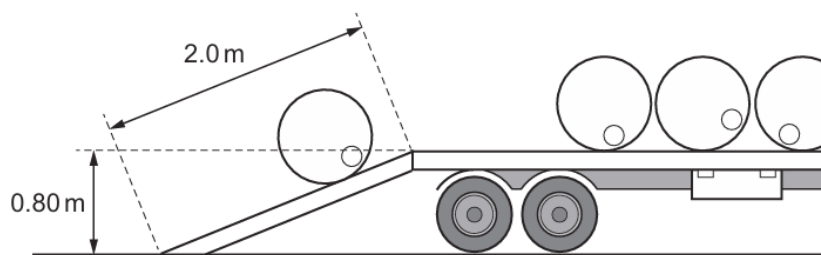


Ignoring the effects of air resistance, which statement about the total energy of the ball is correct?

- A** It is the same at all points.
- B** It is greatest at point 1.
- C** It is greatest at point 2.
- D** It is greatest at point 3.

36. O/N/20/11/Q18

A workman rolls a barrel of weight 2000 N up a plank of length 2.0 m and onto a lorry. The back of the lorry is 0.80 m above the horizontal surface of the road.



What is the work done on the barrel against gravity?

- A** 1000 J
- B** 1600 J
- C** 2500 J
- D** 4000 J

37. O/N/20/11/Q19

Data for three types of electricity generator are shown.

	input energy / MJ	wasted energy / MJ
oil	500	300
nuclear	200	160
hydroelectric	10	1.0

Which is the least efficient generator and which is the most efficient?

	least efficient	most efficient
A	hydroelectric	nuclear
B	hydroelectric	oil
C	nuclear	hydroelectric
D	oil	hydroelectric

38. O/N/20/12/Q14

A student uses an average force of F to push a box a distance of d across a horizontal table. The mass of the box is m and the gravitational field strength is g .

How is the work done by the student pushing the box calculated?

- A** $F \times d$ **B** $\frac{F \times d}{g}$ **C** $m \times d$ **D** $m \times g \times d$

39. O/N/20/12/Q16

What is *power*?

- A** force per second
B the maximum force an object exerts
C the total energy of an object
D work done per second

40. M/J/20/11/Q12

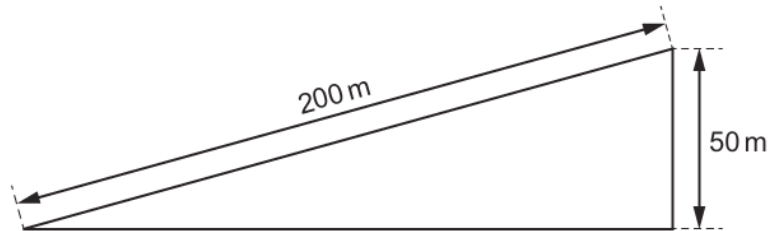
A rocket of mass M when empty carries a mass M of fuel. The rocket and fuel travel at speed v . The engine of the rocket is fired and all of the fuel is expelled. The speed of the rocket increases to $2v$.

What happens to the kinetic energy of the rocket?

- A** It doubles.
B It halves.
C It increases by a factor of four.
D It stays the same.

41. M/J/20/11/Q13

A car of mass 1000 kg is driven 200 m up an incline so that it rises 50 m vertically.



The acceleration of free fall g is 10 m/s^2 .

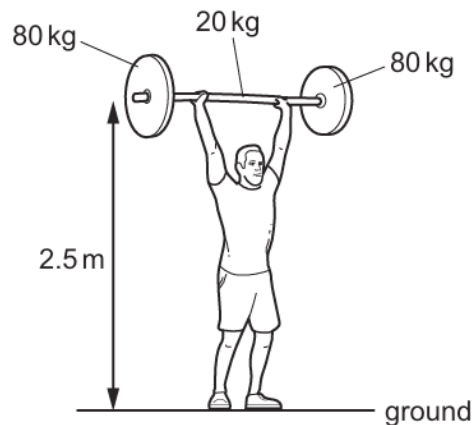
What is the gain in gravitational potential energy?

- A** 5000 J **B** 200 000 J **C** 500 000 J **D** 2 000 000 J

42. M/J/20/11/Q16

In a weightlifting contest, an athlete lifts a metal bar of mass 20 kg fitted with a mass of 80 kg on each end.

The lift from ground to a height of 2.5 m takes 0.50 s.



The gravitational field strength g is 10 N/kg .

What average power does the weightlifter exert in providing the gravitational potential energy during this lift?

- A** 0.90 kW **B** 5.0 kW **C** 8.0 kW **D** 9.0 kW

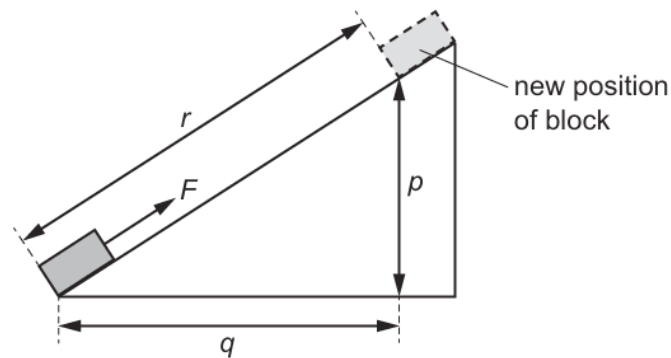
43. M/J/20/11/Q18

In which situation is energy being released by the fusion of hydrogen nuclei to form helium?

- A** in the decay of ^{14}C used to date an object
B in a radioactive isotope emitting alpha-particles
C in the centre of the Earth
D in the centre of the Sun

44. M/J/20/11/Q19

The diagram shows a block being pulled up a slope by a force F .



The block reaches the new position at the top of the slope.

What is the work done by force F in moving the block to its new position?

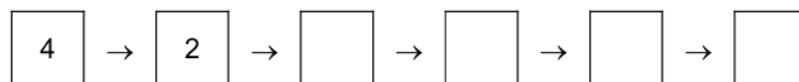
- A** $F \times r$ **B** $F \times p$ **C** $F(q + p)$ **D** $F(r + p)$

45. M/J/20/12/Q19

The main stages in the operation of a coal-fired power station are listed. They are **not** in the correct order.

- 1 The turbine turns a generator.
- 2 Water in the boiler becomes hot.
- 3 Steam turns a turbine.
- 4 Burning coal produces thermal energy.
- 5 Electromagnetic induction produces electrical energy.
- 6 Steam is produced.

The flow chart shows the first two stages.



What is the correct order for the remaining stages?

- A** $6 \rightarrow 3 \rightarrow 1 \rightarrow 5$
B $6 \rightarrow 1 \rightarrow 3 \rightarrow 5$
C $1 \rightarrow 6 \rightarrow 3 \rightarrow 5$
D $6 \rightarrow 3 \rightarrow 5 \rightarrow 1$

46. M/J/20/12/Q20

In which situation is there **no** transfer of energy?

- A a car battery turning the starter motor
- B a car moving along a horizontal road at constant speed
- C a solar panel warming water
- D a spacecraft orbiting the earth at constant height

47. M/J/20/12/Q21

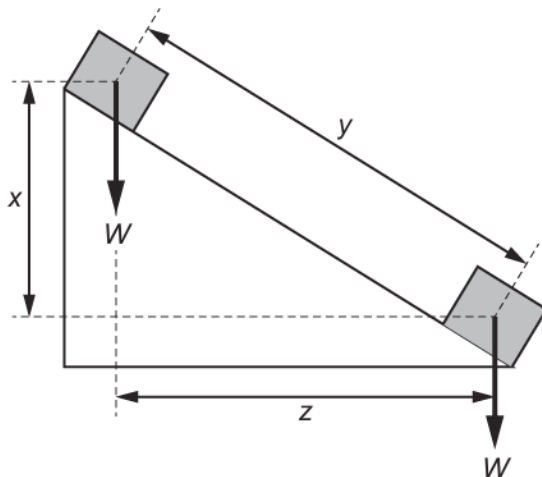
The 200 V battery of an electric car is charged with a current of 33 A for 4.0 hours. The efficiency of the charging process is 90%.

How much useable energy is transferred to the battery?

- A 21 MJ B 86 MJ C 95 MJ D 106 MJ

48. O/N/19/11/Q17

A heavy box of weight W slides down a frictionless slope.



What is the work done by the weight W in moving from top to bottom?

- A Wx B Wy C $W(y - z)$ D $W(x + z)$

49. O/N/19/11/Q18

Where is energy released by the fusion of hydrogen nuclei to form helium?

- A in a nuclear power station
- B in a radioactive isotope emitting alpha-particles
- C in the core of the Earth
- D in the core of the Sun

50. O/N/19/12/Q14

An object X has mass m and velocity v .

A second object Y has mass $2m$ and velocity $2v$.

What is the value of the ratio $\frac{\text{kinetic energy of Y}}{\text{kinetic energy of X}}$?

- A** 1 **B** 2 **C** 4 **D** 8

51. O/N/19/12/Q16

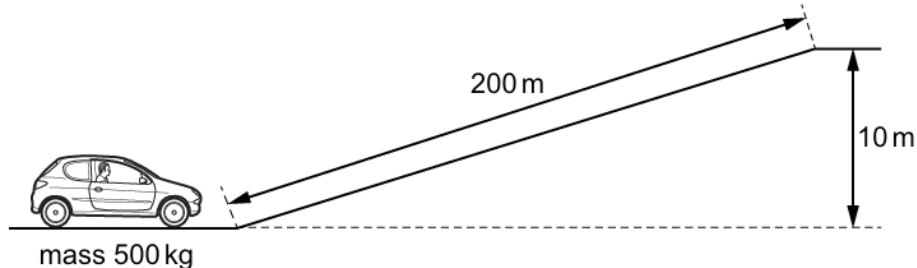
A student uses a newton meter to pull an object a distance d horizontally along a flat surface in a time of t . The newton meter reading is F .

What is given by the expression $(F \times d)/t$?

- A** efficiency
B energy
C power
D work

52. M/J/19/11/Q15

The diagram shows a small car of mass 500 kg approaching a hill. It moves up the hill with constant speed.



The gravitational field strength g is 10 N/kg .

Ignoring friction, how much work is done in moving the car up the hill?

- A** $5.0 \times 10^3 \text{ J}$ **B** $5.0 \times 10^4 \text{ J}$ **C** $1.0 \times 10^5 \text{ J}$ **D** $1.0 \times 10^6 \text{ J}$

53. M/J/19/12/Q16

An object is lifted.

Which expression gives the increase in gravitational potential energy of the object?

- A** mass multiplied by its velocity
B mass multiplied by its acceleration
C the rate at which work is done
D weight multiplied by the increase in height

54. M/J/19/12/Q18

Which method of producing electricity does **not** use an a.c. generator?

- A** geothermal power station
- B** solar panels
- C** wave-energy generators
- D** wind turbine

55. M/J/19/12/Q19

Three solar panels have a total area of 7.2 m^2 . Together, they produce 1.0 kW of power.

An area of 1.0 m^2 of each panel receives 0.80 kJ of energy from the Sun in 1.0 s .

What is the efficiency of the solar panels?

- A** 11% **B** 17% **C** 52% **D** 80%

56. M/J/19/12/Q20

A student suggests three different definitions for power.

- 1 energy transferred per second
- 2 force $\times$ distance moved
- 3 rate of doing work

Which definitions are correct?

- A** 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 3 only

57. O/N/18/11/Q9

An electric motor, connected to the mains, is used to lift bricks to the top of a building.

There is a fuse in the plug.

How can the efficiency of the motor be increased?

- A** Increase the friction in the motor.
- B** Reduce the energy losses in the motor.
- C** Use a fuse with a higher current rating.
- D** Use a fuse with a lower current rating.

58. O/N/18/12/Q8

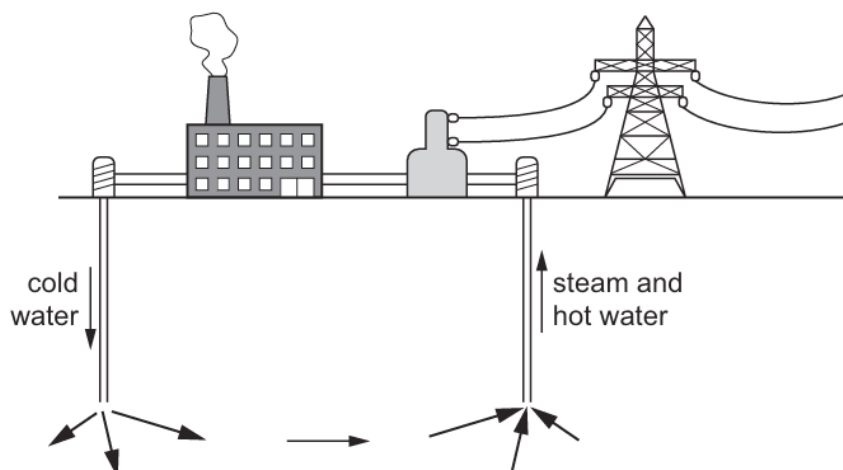
A tennis ball is compressed. Work is done and energy is transferred to the molecules of air inside the ball, and also to the rubber from which the ball is made.

Which row is correct?

	kinetic energy of air molecules inside the ball	temperature of ball
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

59. O/N/18/12/O11

The diagram represents a geothermal power station.

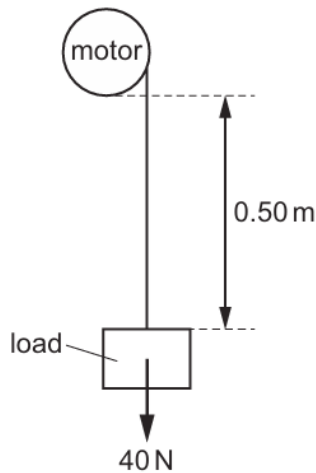


Which useful energy transformation is taking place?

- A** electrical energy → potential energy
- B** electrical energy → thermal energy
- C** potential energy → electrical energy
- D** thermal energy → electrical energy

60. O/N/18/12/Q12

A motor is used to lift a load 0.50 m vertically, as shown.



The load weighs 40 N. The power of the motor is 20 W and the system is 25% efficient.

How long does it take to raise the load?

- A** 0.040 s **B** 0.25 s **C** 4.0 s **D** 40 s

61. M/J/18/11/Q12

A student calculates his power in running up a flight of stairs. He measures the vertical height of the stairs, the time taken to run up the stairs and his weight.

How does he calculate his power?

- A** height $\times$ time $\times$ weight
B $\frac{\text{height} \times \text{weight}}{\text{time}}$
C $\frac{\text{time} \times \text{weight}}{\text{height}}$
D $\frac{\text{weight}}{\text{height} \times \text{time}}$

62. M/J/18/11/Q13

How does an oil-fired power station differ from a nuclear power station?

- A** Gases emitted by hot fuel are emitted into the atmosphere.
B Steam is produced in a boiler using hot fuel.
C The hot steam is used to turn a turbine.
D Turbines are used to drive an electric generator.

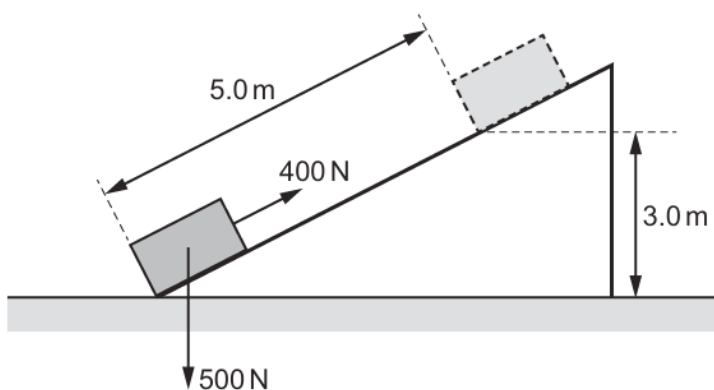
63. M/J/18/12/Q14

Which power station produces carbon dioxide when operating?

- A** gas-fired power station
- B** geothermal power station
- C** nuclear power station
- D** wind power station

64. M/J/18/12/Q15

Work is done when a force of 400 N pulls a crate of weight 500 N at a constant speed along a ramp, as shown.



Part of the work done increases the gravitational potential energy E of the crate and the rest is work done W against friction.

What are the values of E and W ?

	E/J	W/J
A	1500	500
B	1500	2000
C	2000	2500
D	3500	500

65. O/N/17/11/Q9

An astronaut on the Moon drops a tool box of mass 3.0 kg. It falls from rest and its kinetic energy as it hits the surface is 0.96 J.

At which speed does the tool box hit the surface of the Moon?

- A** 0.46 m/s
- B** 0.57 m/s
- C** 0.64 m/s
- D** 0.80 m/s

66. O/N/17/11/Q10

A car travels a distance of 200 m in 20 s. The engine of the car provides a driving force of 1000 N.

What is the power output of the engine?

- A** 0.25 W **B** 4.0 W **C** 100 W **D** 10 000 W

67. O/N/17/12/Q8

A tennis ball of mass 56 g is travelling at 1500 metres / minute.

Which expression is equal to the kinetic energy, in joules, of the tennis ball?

- A** $\frac{1}{2} \times 0.056 \times (25)^2$
B $\frac{1}{2} \times 0.056 \times (1500)^2$
C $\frac{1}{2} \times 56 \times (25)^2$
D $\frac{1}{2} \times 56 \times (1500)^2$

68. O/N/17/12/Q9

In a hydroelectric power station, 4.2×10^5 kg of water passes through the turbines every second. The turbines are at a height of 50 m below the surface of the reservoir. The gravitational field strength g is 10 N/kg.

Assuming there are no energy losses, what is the power output of the power station?

- A** 8.4×10^3 W **B** 8.4×10^4 W **C** 2.1×10^7 W **D** 2.1×10^8 W

69. M/J/17/11/Q15

A small hydroelectric power station diverts water from a river. Every second, 20 kg of water flows through a pipe and falls through a vertical drop of 15 m. The efficiency of the power station is 0.60 (60%).

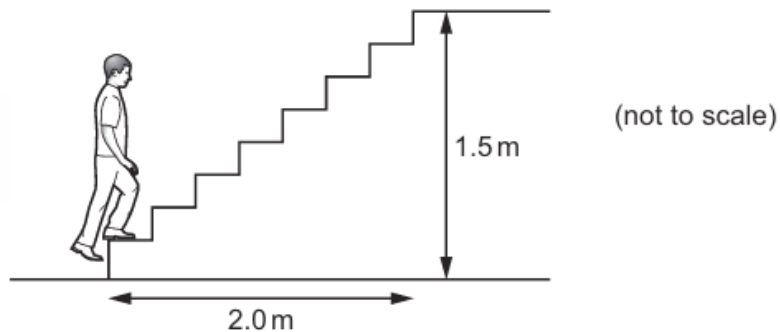
The gravitational field strength g is 10 N/kg.

What is the power output?

- A** 0.18 kW **B** 1.8 kW **C** 3.0 kW **D** 180 kW

70. M/J/17/12/Q15

A student of mass 60 kg climbs some steps. He travels a horizontal distance of 2.0 m and a vertical distance of 1.5 m. The gravitational field strength g is 10 N/kg.



What is the work done against the force of gravity?

- A** 90 J **B** 120 J **C** 900 J **D** 1200 J

71. M/J/17/12/Q16

Which energy source is non-renewable?

- A** nuclear energy
B solar energy
C tidal energy
D wind energy

72. M/J/17/12/Q7

In a coal-fired power station, the coal is burnt and thermal energy (heat) is produced. The thermal energy is used to produce electrical energy.

In which order does the energy pass through parts of the power station?

- A** boiler → generator → turbine
B boiler → turbine → generator
C turbine → boiler → generator
D turbine → generator → boiler

73. O/N/16/11/Q9

Three objects P, Q and R have different masses and different speeds, as shown in the table.

	$\frac{\text{mass}}{\text{kg}}$	$\frac{\text{speed}}{\text{m/s}}$
P	1.0	3.0
Q	2.0	2.0
R	5.0	1.0

What is the order of increasing kinetic energy (smallest first) of the objects?

A P → Q → R

B P → R → Q

C R → P → Q

D R → Q → P

74. O/N/16/11/Q10

A weightlifter lifts some masses through a height of 1.8 m.



The gravitational field strength g is 10 N/kg.

During the lift, the gravitational potential energy of the masses increases by 4500 J.

What is the total mass lifted?

A 25 kg

B 250 kg

C 450 kg

D 810 kg

75. O/N/16/11/Q11

Which expression is used to calculate power?

A $\frac{\text{distance}}{\text{work done}}$

B work done $\times$ distance

C $\frac{\text{work done}}{\text{time taken}}$

D work done $\times$ time taken

76. O/N/16/12/Q12

Which power stations all rely on renewable sources of energy?

- A hydroelectric, oil-fired, solar power stations
- B hydroelectric, solar, wind power stations
- C hydroelectric, oil-fired, wind power stations
- D oil-fired, wind, solar power stations

77. O/N/16/12/Q13

Which force does no work?

- A the force from a car's engine that causes the car to accelerate
- B the force that causes a box to slide over a rough horizontal surface
- C the force that causes a satellite to move in a circular orbit
- D the force that causes a spring to stretch

78. O/N/16/12/Q15

An electric motor is used to operate a lift (elevator). The motor does 140 000 J of work in 70 s.

What is the output power of the motor?

- A 0.50 milliwatts
- B 2.0 watts
- C 2.0 kilowatts
- D 9.8 megawatts

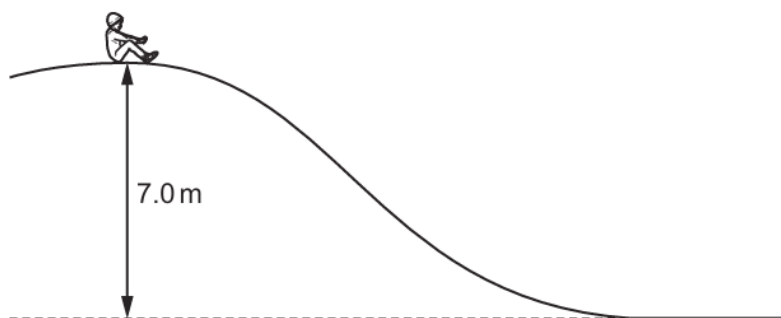
79. O/N/16/12/Q16

Which expression gives the efficiency of an a.c. generator?

- A $\frac{\text{electrical energy output}}{\text{total energy input}}$
- B $\frac{\text{electrical energy output}}{\text{thermal energy output}}$
- C $\frac{\text{total energy output}}{\text{electrical energy input}}$
- D $\frac{\text{thermal energy output}}{\text{total energy input}}$

80. M/J/16/11/Q11

A child slides down a slide.



The weight of the child is 250 N. The height of the slide is 7.0 m. The work done against friction as the child travels down the slide is 1300 J.

What is the change in gravitational potential energy and what is the final kinetic energy of the child?

	change in gravitational potential energy	final kinetic energy
A	1750	450
B	1750	1750
C	17 500	16 200
D	17 500	17 500

81. M/J/16/11/Q12

What uses non-renewable energy?

- A** a geothermal heating system
- B** a nuclear power station
- C** a solar panel
- D** a wind turbine

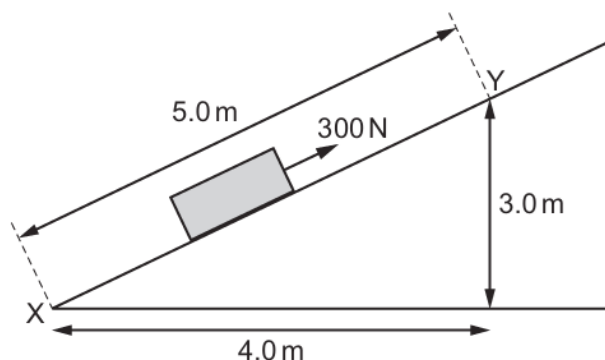
82. M/J/16/11/Q13

Which process in the Sun produces energy?

- A** burning
- B** nuclear fission
- C** nuclear fusion
- D** radiation

83. M/J/16/11/Q14

A 300 N force is applied to a box to move it up a ramp, as shown.



How much work is done by the force when moving the box from X to Y?

- A** 900 J **B** 1200 J **C** 1500 J **D** 3000 J

84. O/N/15/11/Q14

The input power to a lamp is 6.0 W. The lamp wastes 2.7 J of energy in 3.0 s.

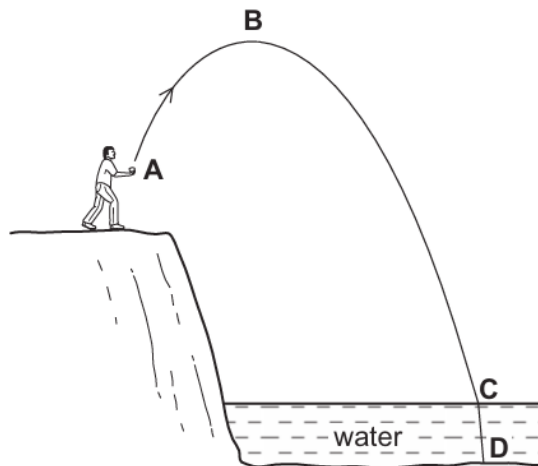
What is the efficiency of the lamp?

- A** 0.15 **B** 0.45 **C** 0.55 **D** 0.85

85. O/N/15/12/Q14

A person throws a stone so that it follows the path shown in the diagram.

In which position does the stone have the most gravitational potential energy?



86. O/N/15/12/Q15

A piston of area 10 cm^2 is pushed slowly into a very large cylinder containing gas at a pressure of 10 N/cm^2 . The pressure of the gas remains constant as the piston moves a distance of 0.10 m .

What is the force of the gas on the piston and what is the work done by the piston on the gas?

	force / N	work done / J
A	1.0	0.1
B	1.0	10
C	100	10
D	100	1000

87. M/J/15/11/Q11

A cyclist travels along a horizontal track at constant speed.

The work done by the cyclist is equal to

- A** the change in kinetic energy.
- B** the force of air resistance.
- C** the force of friction in the bicycle.
- D** the thermal energy (heat) produced.

88. M/J/15/11/Q12

A 2.0 kg mass has 300 J of kinetic energy.

What is the speed of the mass?

- A** 8.7 m/s **B** 12 m/s **C** 17 m/s **D** 24 m/s

89. M/J/15/12/Q12

A rocket of total mass M is travelling at a speed v . The engine of the rocket is fired and fuel is used up. The mass of the rocket decreases to $M/2$ and its speed increases to $2v$.

What happens to the kinetic energy of the rocket?

- A** it doubles
- B** it halves
- C** it increases by a factor of four
- D** it stays the same

90. M/J/15/12/Q13

A builder lifts eight slabs from the ground on to the back of a lorry 1.5 m high.

The total time taken is 48 s and each slab weighs 200 N.

How much useful power does the builder produce?

- A** 50 W **B** 400 W **C** 2400 W **D** 3200 W

91. O/N/14/11/Q13

A student uses a newton meter to pull an object horizontally along rough ground. The student multiplies the newton meter reading by the distance moved by the object.

What does this calculation give?

- A** the efficiency of the process
B the kinetic energy of the object
C the power generated
D the work done on the object

92. O/N/14/11/Q14

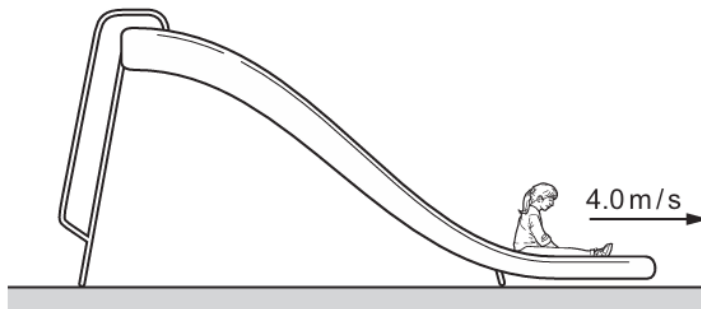
A builder carrying six bricks climbs a ladder. Each brick weighs 25 N and the builder climbs a vertical distance of 4.0 m in 8.0 s.

What is the average power used to raise the bricks?

- A** 13 W **B** 75 W **C** 750 W **D** 4800 W

93. O/N/14/12/Q9

A child of mass 30 kg is moving at a speed of 4.0 m/s when she reaches the bottom of a slide.



What is her kinetic energy?

- A** 60 J **B** 120 J **C** 240 J **D** 480 J

94. M/J/14/11/Q16

A crane lifts a load of 6000 N through a vertical distance of 15 m in 30 s.

What is the average useful power during this operation?

- A** 200 W **B** 400 W **C** 3000 W **D** 12 000 W

95. M/J/14/11/Q17

A javelin has a mass of 0.80 kg.

The javelin is released with 250 J of kinetic energy.

What is the speed of the javelin?

- A** 13 m/s **B** 18 m/s **C** 25 m/s **D** 630 m/s

96. M/J/14/12/Q16

Which source releases carbon dioxide, a greenhouse gas, when generating electricity?

- A** fossil fuels
B geothermal
C hydroelectric
D nuclear

97. O/N/13/11/Q12

Which energy resource comes from hot rocks beneath the Earth's surface?

- A** geothermal energy
B hydroelectric energy
C solar energy
D wind energy

98. O/N/13/12/13

When exposed to light, a solar cell generates electrical energy.

Two solar cells of equal area are tested. Each cell is exposed to sunlight of different brightness for the same time.

Which statement could apply to the more efficient cell?

- A** It generates less electrical energy from light of greater brightness.
B It generates less electrical energy from light of the same brightness.
C It generates the same electrical energy from light of greater brightness.
D It generates the same electrical energy from light of lower brightness.

99. M/J/13/11/Q10

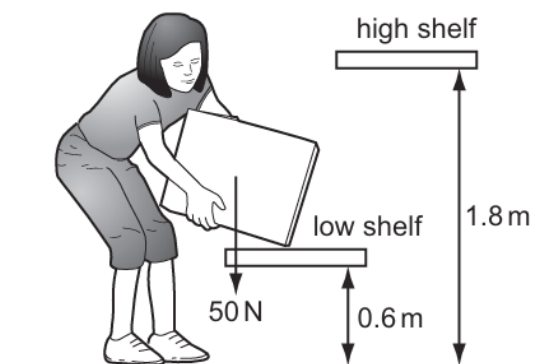
Energy is released in some power stations and in the Sun by either nuclear fission or nuclear fusion.

Which type of nuclear reaction applies in each case?

	power station	Sun
A	fission	fission
B	fission	fusion
C	fusion	fission
D	fusion	fusion

100. M/J/13/11/Q11

The diagram shows a girl lifting a box of weight 50 N from a low shelf to a high shelf.

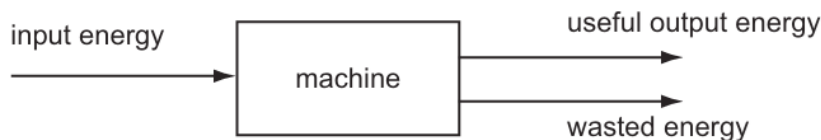


How much work is done on the box?

- A** 30 J **B** 60 J **C** 90 J **D** 120 J

101. M/J/13/11/Q12

The diagram shows the energy transfer through a machine.



The machine is 50% efficient.

Which equation is correct?

- A** input energy = useful output energy
B useful output energy = input energy + wasted energy
C wasted energy = input energy + useful output energy
D wasted energy = useful output energy

102. M/J/13/12/Q13

A swimmer dives into a very deep pool at high speed. He slows down as he moves towards the bottom of the pool.

What is the overall energy transformation as the diver moves downwards through the water?

- A** gravitational potential energy → kinetic energy + thermal energy (heat)
- B** gravitational potential energy → kinetic energy → thermal energy (heat)
- C** kinetic energy + gravitational potential energy → thermal energy (heat)
- D** kinetic energy → gravitational potential energy + thermal energy (heat)

103. M/J/13/12/Q14

A lorry of mass 10 000 kg takes 5000 kg of sand to the top of a hill 50 m high, unloads the sand and then returns to the bottom of the hill.

The gravitational field strength is 10 N/kg.

What is the overall gain in potential energy?

- A** 250 000 J **B** 750 000 J **C** 2 500 000 J **D** 7 500 000 J

104. O/N/12/11/Q13

Which is a non-renewable source of energy?

- A** a battery in a watch
- B** a hydroelectric power station
- C** a solar panel
- D** a wind turbine

105. O/N/12/11/Q14

A builder lifts eight slabs from the ground on to the back of a lorry 1.5 m high.

The total time taken is 48 s and each slab weighs 20 N.

How much useful power does the builder produce?

- A** 5 W **B** 40 W **C** 240 W **D** 320 W

106. O/N/12/12/Q13

Hydroelectric, tidal and fossil fuel power stations generate electrical energy.

Do these use renewable sources of energy?

	hydroelectric	tidal	fossil fuel
A	no	yes	yes
B	no	no	yes
C	yes	no	no
D	yes	yes	no

107. O/N/12/12/Q14

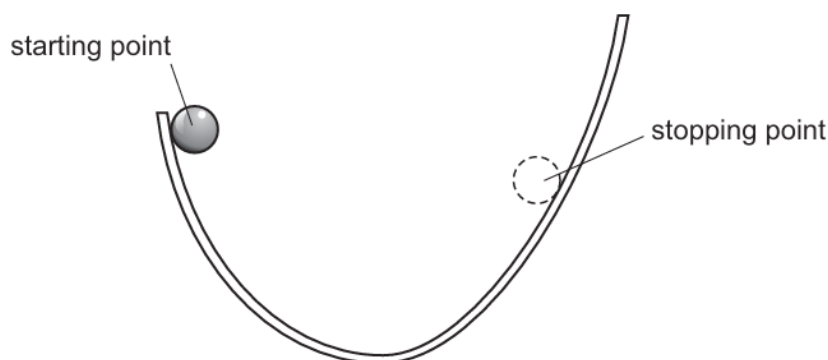
An object of mass 5 kg is carried a distance of 4 m across a room at a constant height above the floor.

What is the work done on the object?

- A** 0 **B** 20 J **C** 50 J **D** 200 J

108. M/J/12/11/Q13

A ball is held at rest on one side of a curved track.



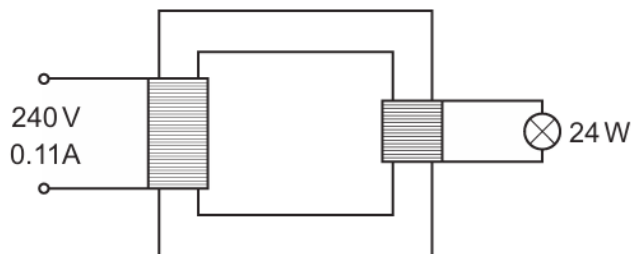
The ball is released. It rolls down one side of the track and part of the way up the other side. It then stops, before rolling back down again. The height of the stopping point is less than that of the starting point.

What is the sequence of energy changes between starting and stopping for the first time?

- A** potential energy → kinetic energy → potential energy
B potential energy → kinetic energy → heat → potential energy
C potential energy → heat → kinetic energy → potential energy
D potential energy → kinetic energy + heat → potential energy + heat

109. M/J/12/11/Q14

A transformer connected to a 240 V mains supply is used to light a 24 W lamp.



The input current to the transformer is 0.11 A and the input voltage is 240 V. The useful output power of the transformer is 24 W.

What is the efficiency of the transformer?

- A** 0.10 **B** 0.91 **C** 1.1 **D** 2.6

110. M/J/12/12/Q13

Two major components of a coal-fired power station are a turbine and a generator.

What are the output forms of energy from the turbine and from the generator?

	turbine	generator
A	electrical	electrical
B	electrical	kinetic
C	heat	kinetic
D	kinetic	electrical

111. M/J/12/12/Q14

What is efficiency?

- A** $\frac{\text{total energy input}}{\text{useful energy output}}$
- B** $\frac{\text{total power input}}{\text{useful energy output}}$
- C** $\frac{\text{useful energy output}}{\text{total energy input}}$
- D** $\frac{\text{useful power output}}{\text{total energy input}}$

112. O/N/11/11/Q10

The efficiency of an electrical generator is 65 %.

The energy input to the generator is 12 kJ.

What is the useful energy output?

- A** 4.2 kJ **B** 5.4 kJ **C** 7.8 kJ **D** 780 kJ

113. O/N/11/11/Q11

Coal is burned as fuel to heat water in a boiler, producing steam. The steam drives a turbine, which is connected to an electric generator.

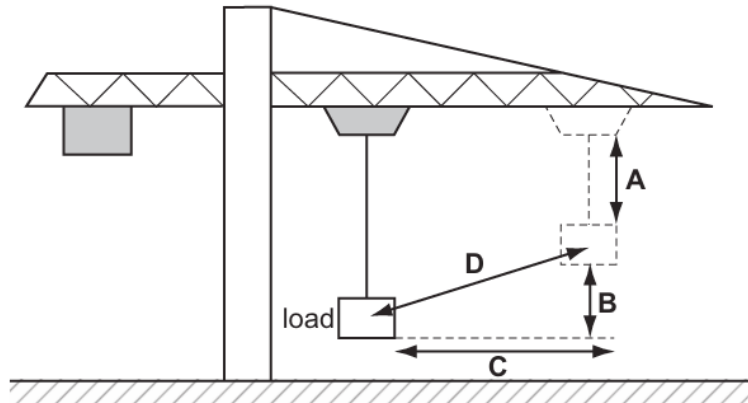
In which order do the major energy transformations take place?

- A** chemical energy → heat energy → electrical energy → kinetic energy
- B** chemical energy → heat energy → kinetic energy → electrical energy
- C** heat energy → chemical energy → electrical energy → kinetic energy
- D** heat energy → chemical energy → kinetic energy → electrical energy

114. O/N/11/11/Q12

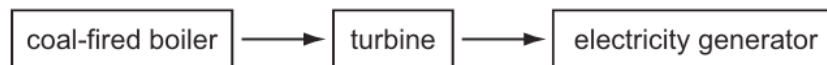
A crane moves its load diagonally, as shown.

By which distance is the weight of the load multiplied to calculate the change in gravitational potential energy of the load?



115. M/J/11/11/Q12

The diagram represents parts of a power station.



What is the order of the energy changes taking place?

- A chemical → heat → kinetic → electrical
- B chemical → kinetic → heat → electrical
- C heat → chemical → kinetic → electrical
- D kinetic → chemical → heat → electrical

116. M/J/11/11/Q13

The centre of the Sun produces large amounts of energy.

What is the source of this energy?

- A chemical reaction
- B nuclear fission
- C nuclear fusion
- D radioactive decay

117. M/J/11/11/Q14

A crane lifts a weight of 1000 N through a vertical height of 30 m.

It uses 60 000 J of energy.

What is the efficiency of the crane?

- A 20 %
- B 30 %
- C 50 %
- D 200 %

118. O/N/10/11/Q15

The energy output of a generator depends on its efficiency and the energy input.

Which set of values is correct?

	efficiency	energy input / MJ	energy output / MJ
A	0.1	50	25
B	0.2	30	6
C	0.4	120	30
D	0.6	60	24

119. M/J/10/11/Q12

A parachutist has opened his parachute and is falling to Earth at constant speed.

What is the principal energy conversion taking place as he falls?

- A** kinetic energy → potential energy
- B** kinetic energy → thermal energy (heat)
- C** potential energy → kinetic energy
- D** potential energy → thermal energy (heat)

120. M./J/10/11/Q13

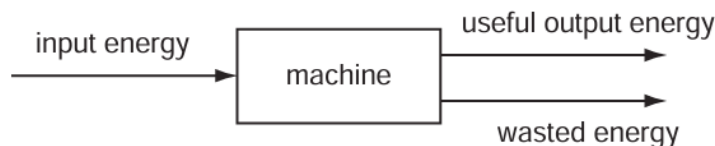
A small emergency generator supplies 432 000 000 J of electrical energy in twenty-four hours.

What is the average power output of the generator?

- A** 5000 W
- B** 300 000 W
- C** 18 000 000 W
- D** 432 000 000 W

121. O/N/09/Q12

The diagram shows energy transfer through a machine.



What is the efficiency of the machine?

- A** $\frac{\text{input energy}}{\text{useful output energy}}$
- B** $\frac{\text{useful output energy}}{\text{input energy}}$
- C** $\frac{\text{useful output energy}}{\text{wasted energy}}$
- D** $\frac{\text{wasted energy}}{\text{input energy}}$

122. O/N/09/Q13

A man weighs 600 N. He runs up a staircase of total height 4.0 metres in 3.0 seconds.

How much useful power is needed to do this?

- A** 450 W **B** 800 W **C** 2400 W **D** 7200 W

123. M/J/09/Q13

A crane lifts a load of 1000 N through a vertical height of 3.0 m in 10 s. The input power to the crane is 500 J/s.

What is the efficiency of the crane?

- A** 0.17 **B** 0.50 **C** 0.60 **D** 0.67

124. O/N/08/Q12

A crane lifts a weight of 1000 N through a vertical height of 30 m.

It uses 60 000 J of energy.

What is the efficiency of the crane?

- A** 20 % **B** 30 % **C** 40 % **D** 50 %

125. O/N/08/Q13

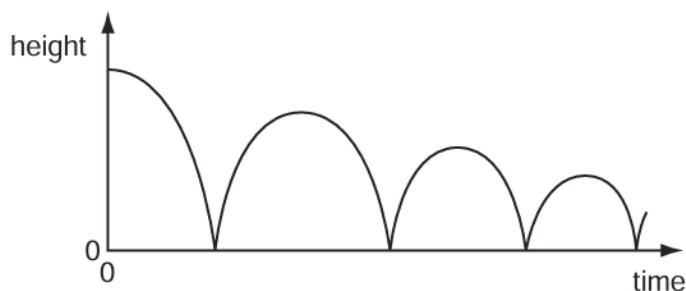
A rock climber of weight 600 N climbs up a rock face of vertical height 300 m in 3600 s.

What is the average power she generates against gravity during this time?

- A** 0.020 W **B** 50 W **C** 1800 W **D** 7200 W

126. O/N/08/Q10

The graph shows how the height above the ground of a bouncing ball changes with time.

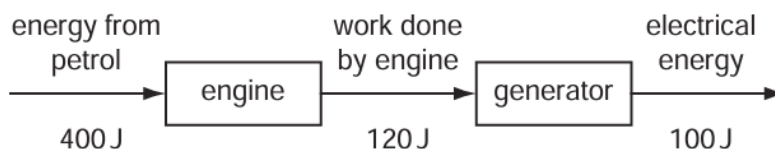


Which statement explains why the height of each peak decreases with time?

- A** Kinetic energy is converted to potential energy at each bounce.
- B** Kinetic energy is converted to thermal energy at each bounce.
- C** The ball gains energy on impact with the floor.
- D** The ball is wearing out.

127. M/J/08/Q11

Energy from petrol is used to operate an engine. The engine drives a generator, which produces electrical energy.



What is the overall efficiency of the process?

- A** 25 %
- B** 30 %
- C** 55 %
- D** 83 %

128. O/N/07/Q9

A 2 kg mass is moving at constant speed.

The kinetic energy of the mass is 400 J.

What is the speed of the mass?

- A** 0.4 m/s
- B** 20 m/s
- C** 200 m/s
- D** 400 m/s

129. O/N/07/Q11

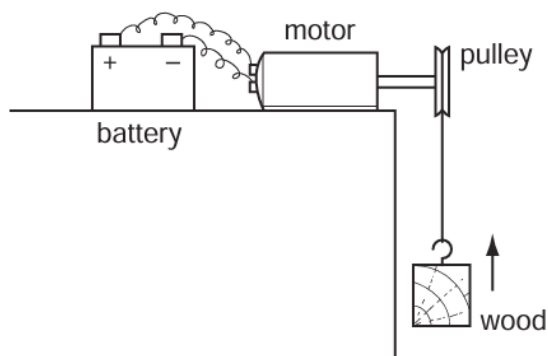
The efficiency of an electrical generator is 65 %.

Which useful output can be expected if the energy input to the generator is 12 kJ?

- A** 4.2 kJ
- B** 7.8 kJ
- C** 19 kJ
- D** 780 kJ

130. M/J/07/Q9

The diagram shows a battery-operated motor lifting a block of wood at constant speed.



What is the overall energy change taking place?

- A** chemical → gravitational potential
- B** gravitational potential → electrical
- C** gravitational potential → kinetic
- D** kinetic → gravitational potential

131. M/J/07/Q10

Four students exercise in a gym.

Which student does the most work?

	exercise time / s	power developed / W
A	50	250
B	100	150
C	200	200
D	250	30

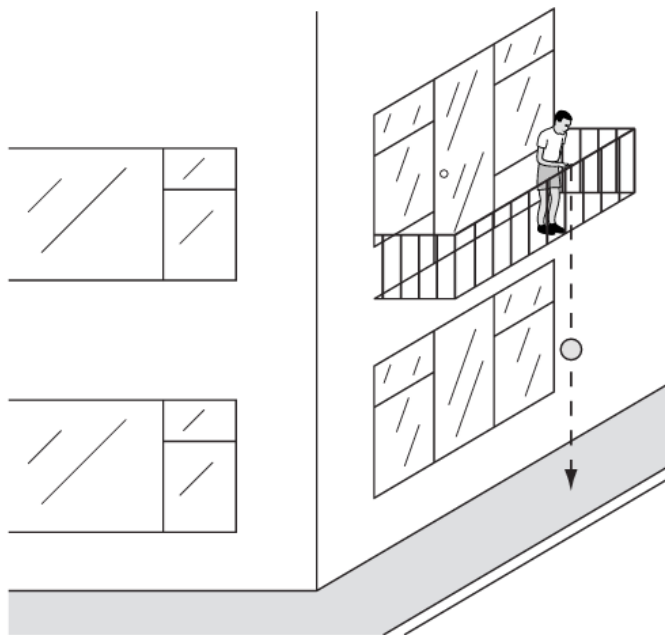
132. M/J/07/Q11

Which formula gives the efficiency of an energy conversion?

- A** efficiency = total energy input - useful energy output
- B** efficiency = useful energy output × total energy input
- C** efficiency = $\frac{\text{useful energy output}}{\text{total energy input}}$
- D** efficiency = $\frac{\text{total energy input}}{\text{useful energy output}}$

133. O/N/06/Q8

A young child holds a ball over the edge of a balcony. The ball has gravitational potential energy. The ball is then released. It falls onto a concrete path below, and bounces back up.



Which sequence represents, in the correct order, the transformations of the gravitational potential energy after the ball is released?

- A → elastic potential energy → kinetic energy → chemical potential energy
- B → elastic potential energy → kinetic energy → gravitational potential energy
- C → kinetic energy → elastic potential energy → kinetic energy
- D → kinetic energy → gravitational potential energy → kinetic energy

134. O/N/06/Q9

How much energy would be released if 1×10^{-20} kg of matter was entirely converted to energy? (The speed of light is 3×10^8 m/s.)

- A 3×10^{-12} J B 9×10^{-7} J C 4.5×10^{-4} J D 9×10^{-4} J

135. M/J/06/Q11

In a hydroelectric power station, water from a reservoir falls down a long pipe before entering the turbines. The turbines then turn the generator.

What is the **overall** energy conversion?

- A electrical energy into kinetic energy
- B electrical energy into potential energy
- C kinetic energy into chemical energy
- D potential energy into electrical energy

136. M/J/06/Q12

Which statement about fission or fusion is correct?

- A** During fission, hydrogen converts into helium and releases energy.
- B** During fission, uranium converts into daughter products and releases energy.
- C** During fusion, helium converts into hydrogen and releases energy.
- D** During fusion, uranium converts into daughter products and releases energy.

137. M/J/06/Q13

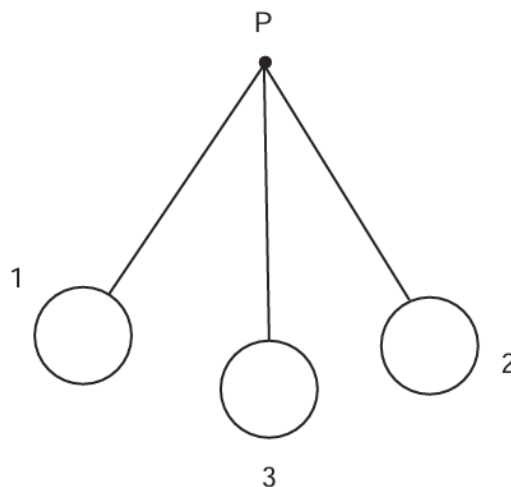
The input power to a motor is 300 W. In 20 s it lifts a load of 400 N through a height of 6.0 m.

What is the efficiency of the motor?

- A** 12 % **B** 25 % **C** 40 % **D** 75 %

138. O/N/05/Q11

A mass hangs on a string fixed at point P. It starts from position 1 and swings to the furthest position on the opposite side, position 2. It then oscillates several times with decreasing amplitude before ending at position 3.

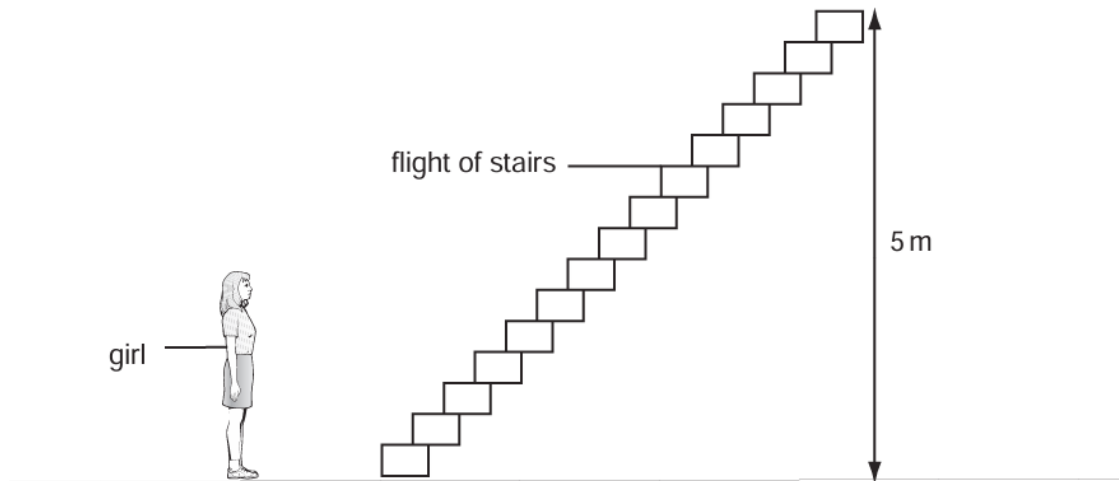


Where does the ball have the most kinetic energy?

- A** at position 1
- B** at position 2
- C** the first time at position 3
- D** the last time at position 3

139. O/N/05/Q12

A girl of weight 500 N runs up a flight of stairs in 10 seconds. The vertical height of the stairs is 5 m.



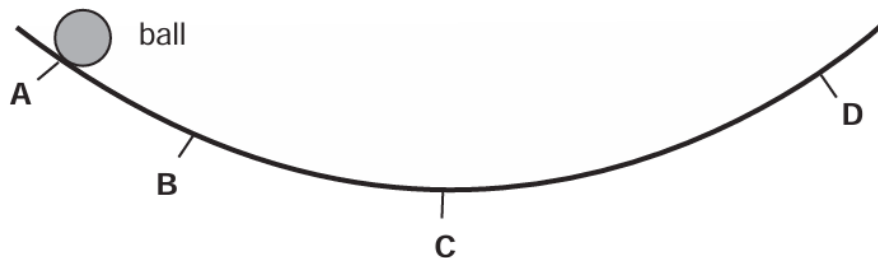
What is the average power developed by the girl?

- A** 50 W **B** 100 W **C** 250 W **D** 1000 W

140. M/J/05/Q10

The diagram shows a curved curtain rail that has a steel ball rolling on it. The ball is released at point A.

At which point does the ball have maximum kinetic energy?



141. M/J/05/Q11

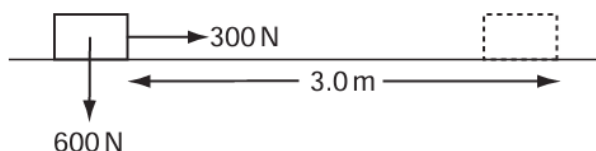
The speed of light is 3×10^8 m/s.

What is the energy equivalent of 2 kg of matter?

- A** $2 \times 3 \times 10^8$ J
B $2000 \times 3 \times 10^8$ J
C $\frac{1}{2} \times 2 \times (3 \times 10^8)^2$ J
D $2 \times (3 \times 10^8)^2$ J

142. O/N/04/Q13

When a 300 N force is applied to a box weighing 600 N, the box moves 3.0 m horizontally in 20 s.



What is the average power?

- A** 45 W **B** 90 W **C** 900 W **D** 1800 W

143. M/J/04/Q11

When a nucleus of Uranium-235 absorbs a neutron, nuclear fission occurs. In a typical reaction the total mass decreases by 3×10^{-28} kg.

Given that the speed of light c is 3×10^8 m/s, approximately how much energy is released?

- A** 9×10^{-20} J
B 2×10^{-13} J
C 3×10^{-11} J
D 3×10^{-5} J

144. M/J/04/Q12

A car is driven along a level road. The total energy input from the petrol is 60 kJ, and the car wastes 45 kJ of energy.



What is the efficiency of the car?

- A** 15 % **B** 25 % **C** 45 % **D** 75 %

145. M/J/04/Q13

Which represents the main energy changes that take place in a coal-fired power station?

- A** chemical $\rightarrow$ heat $\rightarrow$ kinetic $\rightarrow$ electrical
B chemical $\rightarrow$ heat $\rightarrow$ light $\rightarrow$ electrical
C chemical $\rightarrow$ kinetic $\rightarrow$ electrical $\rightarrow$ potential
D kinetic $\rightarrow$ heat $\rightarrow$ light $\rightarrow$ electrical